

Exp No:5b

Data Visualization using Matplotlib

Date:

1. Box Plot

Aim

To visualize the distribution and identify the spread and possible outliers in Likes, Comments, Shares, and Views using a box plot

Algorithm

1. Import Pandas and Matplotlib libraries and load the dataset.
2. Select the numerical columns Likes, Comments, Shares, and Views.
3. Create a box plot using Matplotlib.
4. Add suitable labels and title and display the plot.

CODE:

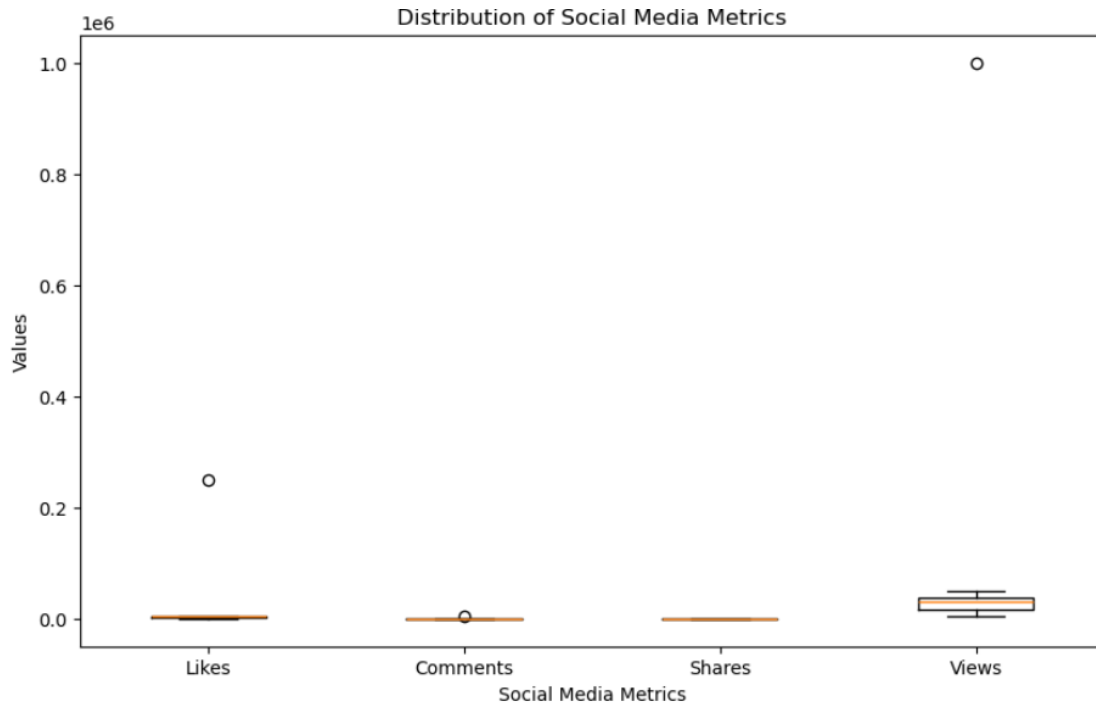
```
# Load dataset
df = pd.read_csv(
    r"C:\Users\SHREE HARINI\Downloads\social_media_dataset_with_null_duplicates_outliers.csv"
)

# Box plot
plt.figure(figsize=(10, 6))

plt.boxplot(
    [df["Likes"].dropna(),
     df["Comments"].dropna(),
     df["Shares"].dropna(),
     df["Views"].dropna()],
    labels=["Likes", "Comments", "Shares", "Views"]
)

plt.xlabel("Social Media Metrics")
plt.ylabel("Values")
plt.title("Distribution of Social Media Metrics")

plt.show()
```



2. Multiple Line Chart

Aim

To compare multiple social media engagement metrics across different posts using multiple line charts in Matplotlib.

Algorithm:

1. Import Pandas and Matplotlib libraries and load the dataset.
2. Select Post_ID, Likes, Comments, and Shares.
3. Plot Likes, Comments, and Shares as separate lines on the same graph.
4. Add labels, title, and legend and display the chart.

Code:

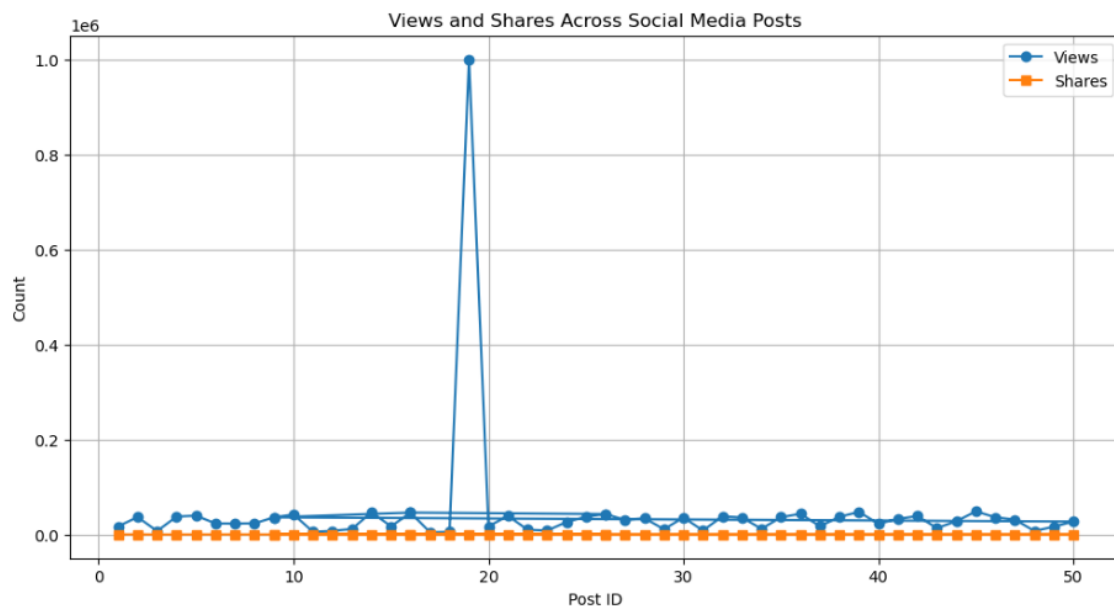
```
df = pd.read_csv(
    r"C:\Users\SHREE HARINI\Downloads\social_media_dataset_with_null_duplicates_outliers.csv"
)

plt.figure(figsize=(12, 6))

plt.plot(df["Post_ID"], df["Views"], marker="o", label="Views")
plt.plot(df["Post_ID"], df["Shares"], marker="s", label="Shares")

plt.xlabel("Post ID")
plt.ylabel("Count")
plt.title("Views and Shares Across Social Media Posts")

plt.legend()
plt.grid(True)
plt.show()
```



RESULT: Thus, the dataset was successfully visualized using box plots and multiple line charts to analyze metric distribution and engagement variations.